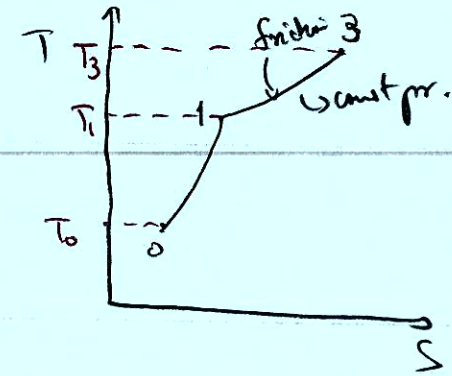
3/28/2019

Shock contribution to temperature increase in inlets

Let '1' represent the conditions @ the end of shock compression

Let '3' " " " " " " " shock + friction



$$\psi = T_3/T_0$$

$$\psi_1 = T_1/T_0$$

- Using the definition of efficiencies

$$\frac{p_1}{p_0} = \left\{ \frac{\psi_1}{\psi_1(1-\eta_1') + \eta_1'} \right\}^{\gamma_c/(\gamma_c-1)}$$

$\eta_1' \equiv$ efficiency as in text book.

$$\frac{p_3}{p_0} = \left\{ \frac{\psi}{\psi(1-\eta_c') + \eta_c'} \right\}^{\frac{\gamma_c}{\gamma_c-1}}$$

$$p_1 = p_3 \Rightarrow \quad \eta_c' = \eta_1' \left(\frac{1 - \frac{1}{\psi_1}}{1 - \frac{1}{\psi}} \right)$$

$$\boxed{\psi_1 = 1 + \frac{\gamma_c - 1}{2} M_0^2 \left(1 - \left(\frac{V_1}{V_0} \right)^2 \right)}$$

$V_1 \equiv$ vel after shock comp. alone.

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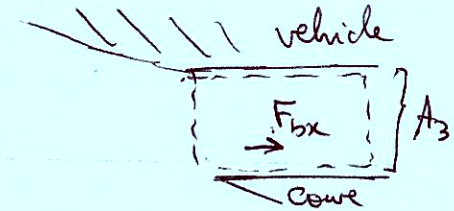
What is $\frac{V_1}{V_0}$?



$$\dot{m}_0 V_1 + F_{bx} = \dot{m}_0 V_3$$

$$F_{bx} = -C_f \rho_3 \frac{V_3^2}{2} A_w = -C_f \frac{\dot{m}_0 V_3}{2} \cdot \frac{A_w}{A_3} \quad (\text{But } \rho_3 V_3 A_3 = \dot{m})$$

$$F_{bx} = -\dot{m}_0 V_3 \left(\frac{C_f}{2} \frac{A_w}{A_3} \right)$$

$$\frac{V_1}{V_3} = 1 + \frac{C_f}{2} \frac{A_w}{A_3}$$

Energy cons. between '0' & '3'

$$\Rightarrow \dot{m}_0 \frac{V_0^2}{2} + C_{p_c} T_0 \dot{m}_0 = \dot{m}_0 \frac{V_3^2}{2} + C_{p_c} T_3 \dot{m}_0$$

$$\left(\frac{V_3}{V_0} \right)^2 = 1 + 2 \left(\frac{C_{p_c} T_0}{V_0^2} - \frac{C_{p_c} T_3}{V_0^2} \right) = 1 + \frac{2(1-\psi)}{(\gamma_c - 1) M_0^2}$$

$$\boxed{\psi_1 = 1 + \frac{\gamma_c - 1}{2} M_0^2 \left\{ 1 - \left[1 + \underbrace{\frac{C_f}{2} \frac{A_w}{A_3}}_{0.01-0.05} \right]^2 \left[1 + \frac{2(1-\psi)}{(\gamma_c - 1) M_0^2} \right] \right\}}$$

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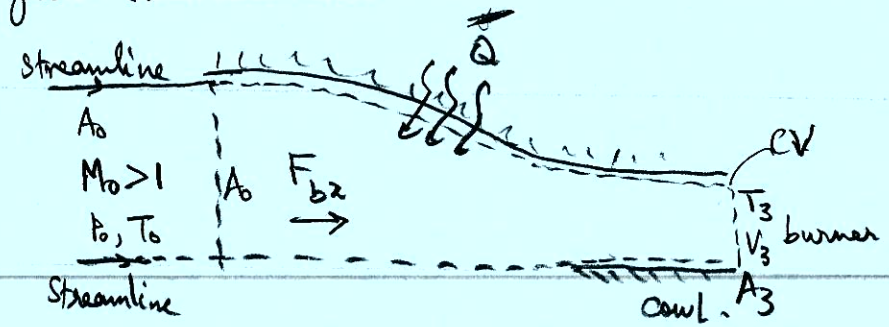
Determining comp. efficiency from global measurements.

F_{bx} : total axial force that includes the pr & skin friction acting along the inlet walls.



Continuity : $\rho_0 u_0 A_0 = \rho_3 u_3 A_3 = \dot{m}_0$ (1)

x Mom : $p_0 A_0 + \dot{m}_0 V_0 + F_{bx} = p_3 A_3 + \dot{m}_0 V_3$ (2)

Energy : $C_{p_c} T_0 + \frac{V_0^2}{2} + \frac{\dot{Q}}{\dot{m}_0} = C_{p_c} T_3 + \frac{V_3^2}{2}$ (3)

$p = \rho R T \Rightarrow p_3 = \rho_3 R_c T_3 = \frac{\dot{m}_0 R_c T_3}{A_3 V_3}$ (4)

$$\rho_3 = \frac{p_3}{R_c T_3}$$

Knowns : ρ_0, u_0, A_0, p_0, Q

Unknowns : V_3, T_3, p_3, ρ_3

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$$V_3 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

'+' $\Rightarrow$ supersonic $V_3 \Rightarrow$ scram

'−' $\Rightarrow$ subsonic $V_3 \Rightarrow$ ramjets

$$a = \frac{\gamma_c + 1}{\gamma_c}$$

$$b = -\left(V_0 + \frac{P_0 A_0 + F_{bx}}{\dot{m}_0} \right)$$

$$c = \frac{\gamma_c - 1}{\gamma_c} \left(C_{p_c} T_0 + \frac{V_0^2}{2} + \frac{\bar{Q}}{\dot{m}_0} \right)$$

Note expression for c in the text book

$$c = R_c \left(C_{p_c} T_0 + \frac{V_0^2}{2} + \frac{Q}{\dot{m}_0} \right)$$ → I think this is wrong.

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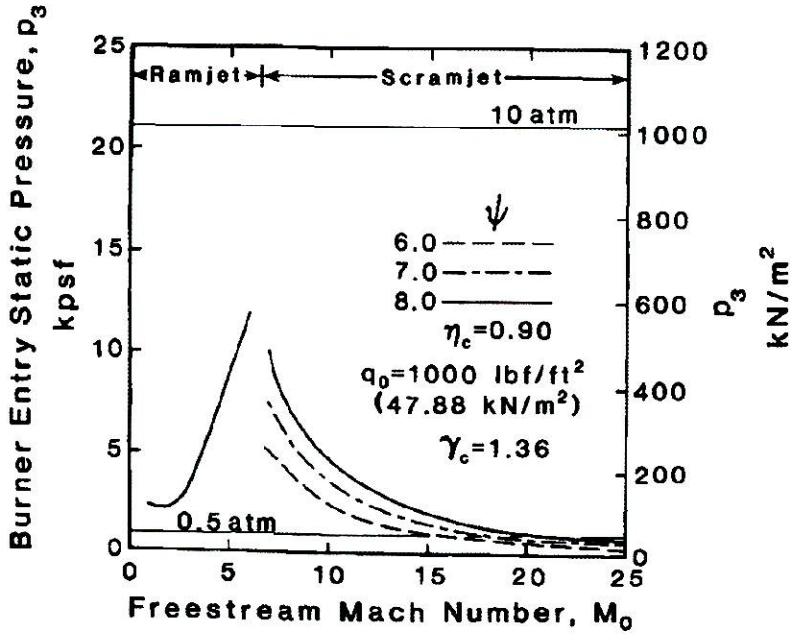

Fig. 5.26 Burner entry static pressure as a function of freestream Mach number and cycle static temperature ratio.

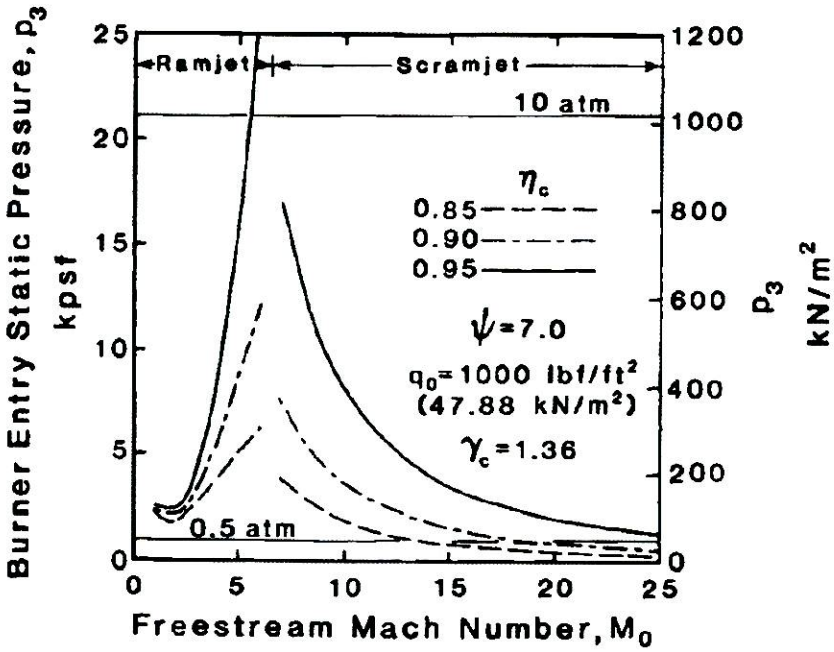

Fig. 5.25 Burner entry static pressure as a function of freestream Mach number and compression efficiency.

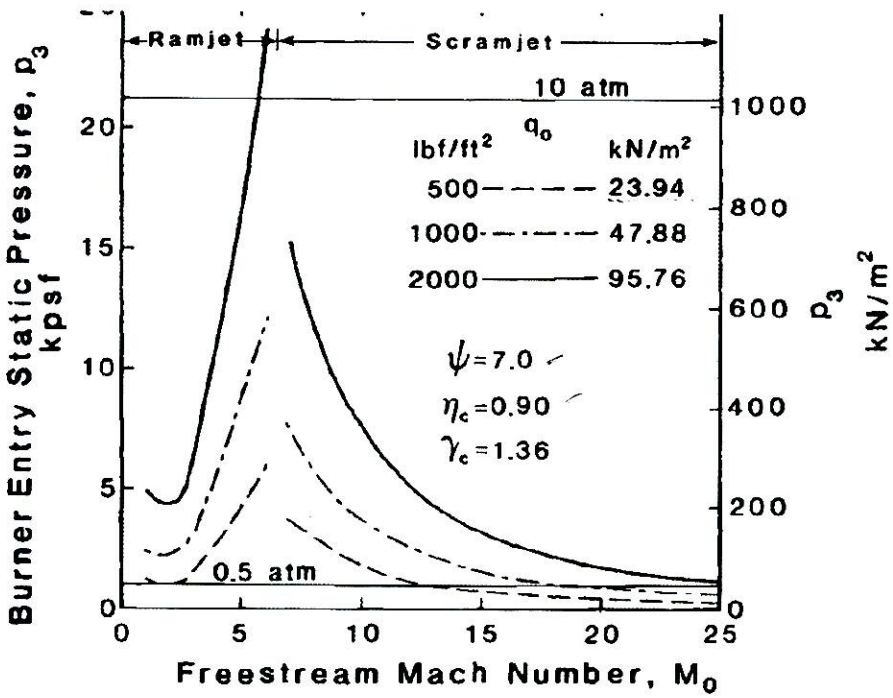

Fig. 5.24 Burner entry static pressure as a function of freestream

Burner Entry pressures.

- p_3 the max pressure anywhere within the engine. So, structural constraints require $p_{3_{max}} = 10$ atm

- For chemical reactions to occur efficiently, $p_{3,min} = 0.5$ atm

- For a given ψ, η_c', M_0

$$\frac{p_3}{p_0} = \left\{ 1 - \eta_c' \left(1 - \frac{1}{\psi} \right) \right\}^{-\frac{\gamma_c}{\gamma_c - 1}}$$

But $p_0 = \dfrac{2q_0}{\gamma_c M_0^2}$ (q_0 : dyn pr, typically - 20-100 kPa)

$$\boxed{p_3 = \frac{2q_0}{\gamma_c M_0^2 \left(1 - \eta_c' \left(1 - \frac{1}{\psi} \right) \right)^{\frac{\gamma_c}{\gamma_c - 1}}}}$$

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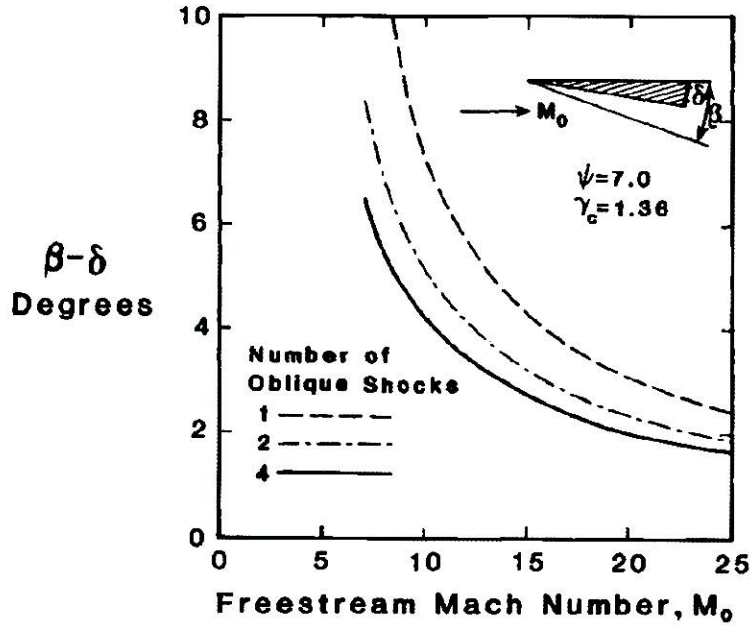

Fig. 5.27 Angle between the initial oblique shock wave and the vehicle compression surface as a function of freestream Mach number.

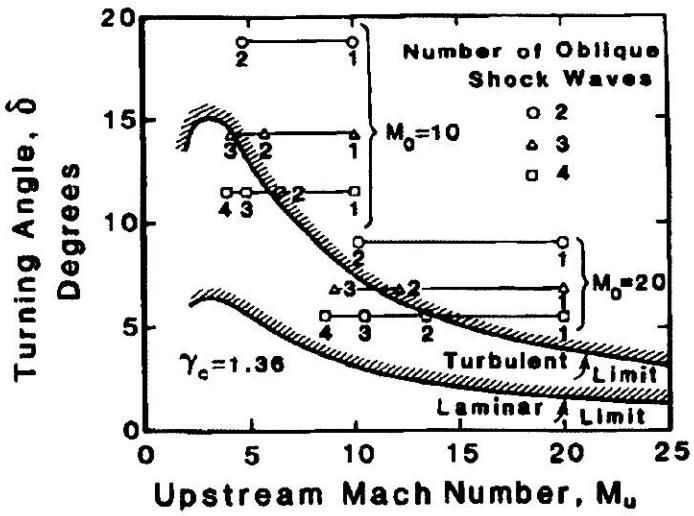

Fig. 5.29 Maximum allowable turning without boundary-layer separation for originating shock waves as a function of upstream Mach number, in accordance with Eqs. (5-35) and (5-36). The discrete symbols are taken from the designs of Sec. 5.5.2 for ψ = 7.0.

Aerodynamics Considerations

- Directly or indirectly affects η_c
- Most considerations come from viscous interaction & flow sep. prevention
- Flow sep causes
  1) pressure drag that is $\gg$ skin friction
  2) unsteadiness that propagate thru the duct
  3) Increased transport of hot gases to wall $\Rightarrow$ hot spots
- Typical rule of thumb for sep

$$\left(\frac{M_d}{M_u}\right)_{lam} < 0.898$$

$$\left(\frac{M_d}{M_u}\right)_{turb} < 0.762$$

M_d : Mach # d/s of a shock
M_u : " " u/s " " "

- Dictates $\frac{p_d}{p_u}$ across the shock & max flow turning angle across the shock.

$\Rightarrow$ Determine # of O.S. reqd.

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Inlet Starting Process

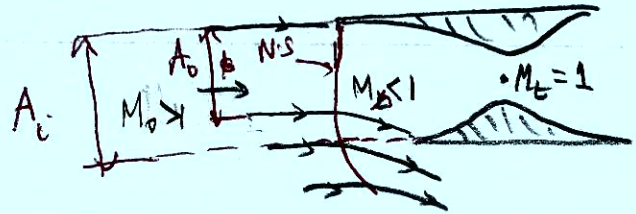
· Subcritical ·

⇒ N.S. in front of inlet

– Inlet is <u>not</u> started

– Spillage issues.

– How do you find spillage $\left(\frac{A_o}{A_i}\right)$ ratio ?



Simple approach } N.S ⇒ M_s after N.S. is known ; we know $M_t = 1$

⇒ $A_t = A^*$

⇒ Given M_s & A^*, isen tables ⇒ A_o

Critical

⇒ – Normal shock @ the face of inlet.

Supercritical

– N.S. is "swallowed" & lives in diverging section close to the throat

– $M_t > 1$

– $\pi_c = f(M_t) \Rightarrow \eta_c = f(\pi_c)$

↑ p_t ratio across shock

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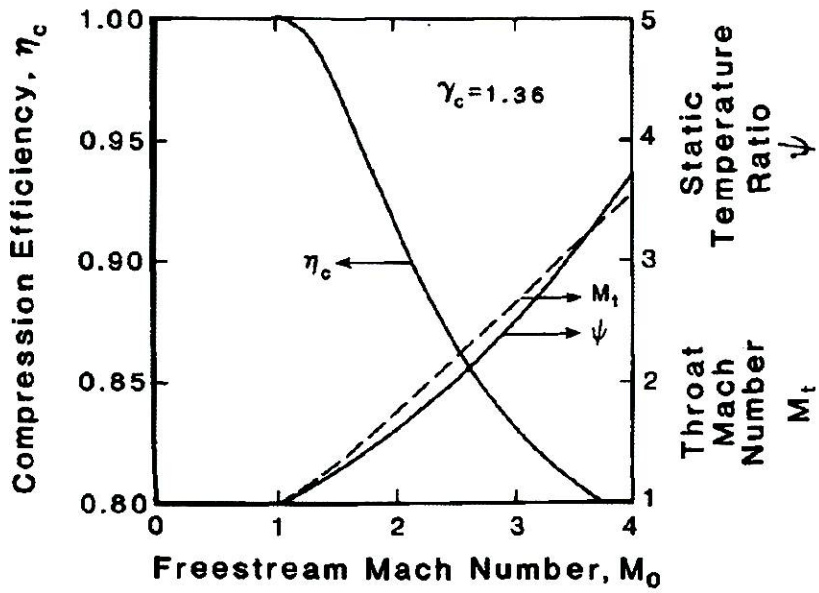

Fig. 5.33 Fixed geometry pitot compression system throat Mach number, static temperature ratio, and adiabatic compression efficiency as functions of the starting freestream Mach number.